POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6021T | Semester VI | Theory | 4 Credits

Refrigeration and Air Conditioning

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6021T as Refrigeration and Air Conditioning in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently based on public diploma notes from Ganesh Institute of Engineering and Technology and the IIT Roorkee NPTEL Refrigeration and Air-conditioning course outline. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Refrigerant charging/recovery, pressure testing, electrical isolation, brazing, leak testing and system commissioning require approved equipment, current regulations, PPE, ventilation and competent supervision.

Verified source note: Verified sources support air-refrigeration and vapour-compression cycles, P-h representation, vapour absorption, components, refrigerants, psychrometry, cooling load, air distribution, thermal comfort and indoor environment. The handbook presents these as four learner-friendly modules rather than an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Refrigeration and Air Conditioning develops the principles used to remove heat, preserve products and provide safe, comfortable indoor conditions. It connects air, vapour-compression and absorption cycles with equipment, refrigerants, psychrometrics, cooling load, air distribution and responsible operation.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□.

Prerequisite foundation

Thermodynamics, heat transfer, basic fluid mechanics, engineering drawing, electrical awareness and laboratory/workshop safety.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain refrigeration terms, air refrigeration and vapour-compression cycle fundamentals.
2. Explain vapour-absorption systems and the functions of refrigeration-system components.
3. Explain refrigerant selection, refrigeration applications and safe environmental practice.
4. Apply psychrometric and comfort-air-conditioning concepts to simple load and system studies.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain refrigeration effect, COP and air/vapour-compression refrigeration-cycle fundamentals.	Understand
CO2	Explain vapour-absorption refrigeration and the construction/function of major refrigeration components.	Understand
CO3	Explain refrigerant properties, flow controls, applications and responsible handling considerations.	Understand
CO4	Interpret moist-air properties and psychrometric processes for comfort air-conditioning and simple cooling-load studies.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Refrigeration Fundamentals and Cycles	Refrigeration effect, COP, units, air-refrigeration concept, reverse-Brayton/Bell–Coleman cycle and basic vapour-compression refrigeration.	Approx. 15 hours	CO1	Module 1
2	Absorption Systems and Refrigeration Equipment	Vapour-absorption principle; compressors, condensers, evaporators, cooling towers and auxiliaries; construction, function and basic performance links.	Approx. 15 hours	CO2	Module 2
3	Refrigerants, Controls and Applications	Expansion devices, refrigerant properties and designation, selection criteria, applications such as cold storage and domestic systems, environmental and safety responsibilities.	Approx. 14 hours	CO3	Module 3
4	Psychrometrics and Comfort Air Conditioning	Moist-air properties, psychrometric chart, heating/cooling/humidification/dehumidification/mixing processes, thermal comfort, simple cooling load and air-distribution awareness.	Approx. 16 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Refrigeration Fundamentals and Cycles

Refrigeration effect, COP, units, air-refrigeration concept, reverse-Brayton/Bell–Coleman cycle and basic vapour-compression refrigeration.

CO1 · Approx. 15 hours

Absorption Systems and Refrigeration Equipment

Vapour-absorption principle; compressors, condensers, evaporators, cooling towers and auxiliaries; construction, function and basic performance links.

CO2 · Approx. 15 hours

Refrigerants, Controls and Applications

Expansion devices, refrigerant properties and designation, selection criteria, applications such as cold storage and domestic systems, environmental and safety responsibilities.

CO3 · Approx. 14 hours

Psychrometrics and Comfort Air Conditioning

Moist-air properties, psychrometric chart, heating/cooling/humidification/dehumidification/mixing processes, thermal comfort, simple cooling load and air-distribution awareness.

CO4 · Approx. 16 hours

MODULE 1

Refrigeration Fundamentals and Cycles

Refrigeration effect, COP, units, air-refrigeration concept, reverse-Brayton/Bell–Coleman cycle and basic vapour-compression refrigeration.

Approx. 15 hours **CO1** **Understand · Apply**

Learning goals

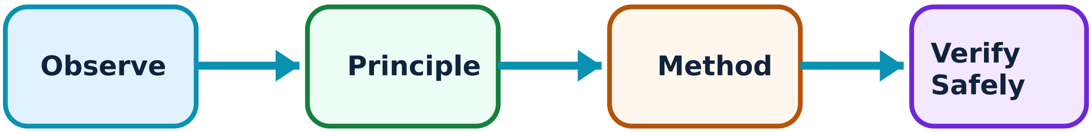
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Refrigeration Fundamentals and Cycles: identify the system, state its principle, follow the correct method and verify the result safely.

1. Refrigeration effect and performance−

Refrigeration removes heat from a controlled low-temperature space and rejects it elsewhere. Refrigerating effect expresses useful cooling; coefficient of performance compares useful cooling to required work. A valid calculation needs a clear system boundary and consistent units.

Study and application method

For a supplied cycle sketch or state data, identify the refrigerated space, heat absorbed, work input, heat rejected and the appropriate COP expression before substituting values.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a supplied cycle sketch or state data, identify the refrigerated space, heat absorbed, work input, heat rejected and the appropriate COP expression before substituting values.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not use classroom COP calculations to predict an installed system without actual load, losses, controls, ambient conditions and component data.

2. Air and vapour-compression cycles−

Air refrigeration commonly uses reverse-Brayton principles. A simple vapour-compression system circulates refrigerant through compressor, condenser, expansion device and evaporator; phase change enables effective heat transfer. T–s and P–h diagrams help visualise state changes.

Study and application method

Trace each component in the cycle, label pressure/temperature trends and identify where work, heat rejection, throttling and refrigerating effect occur.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Pressure and temperature conditions can be hazardous. Never open, charge or modify a circuit as a learning exercise without authorisation and certified equipment.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Absorption Systems and Refrigeration Equipment

Vapour-absorption principle; compressors, condensers, evaporators, cooling towers and auxiliaries; construction, function and basic performance links.

Approx. 15 hours | **CO2** | **Understand · Apply**

Learning goals

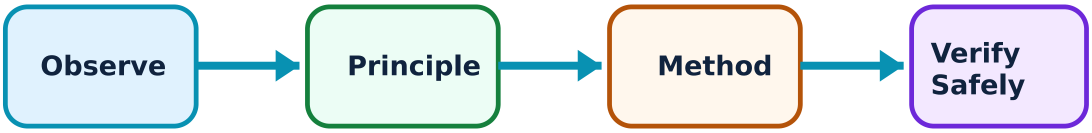
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Absorption Systems and Refrigeration Equipment: identify the system, state its principle, follow the correct method and verify the result safely.

1. Vapour-absorption refrigeration−

Absorption systems use a heat-driven generator/absorber arrangement in place of a mechanical compressor for part of the compression function. Their application depends on available heat, working-pair choice, cooling-water needs, scale and efficiency considerations.

Study and application method

Draw the conceptual absorber–pump–generator–condenser–expansion–evaporator path and distinguish heat input, heat rejection, solution circulation and refrigerant circulation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw the conceptual absorber–pump–generator–condenser–expansion–evaporator path and distinguish heat input, heat rejection, solution circulation and refrigerant circulation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Working fluids, heat sources and pressurised equipment require system-specific knowledge; do not assume different absorption pairs have interchangeable hazards or maintenance needs.

2. Major components and heat rejection−

Compressors raise vapour pressure; condensers reject heat; expansion devices meter/pressure-reduce refrigerant; evaporators absorb heat. Air- and water-cooled condensers, cooling towers and evaporator types are selected according to duty, environment, water availability and maintenance capability.

Study and application method

For each component, state inlet/outlet condition trends, primary function, likely fault symptom and a non-invasive inspection item.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For each component, state inlet/outlet condition trends, primary function, likely fault symptom and a non-invasive inspection item.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrical drives, moving fans, hot surfaces, water-treatment chemicals and high-pressure parts require lockout, guarding and approved maintenance practice.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Refrigerants, Controls and Applications

Expansion devices, refrigerant properties and designation, selection criteria, applications such as cold storage and domestic systems, environmental and safety responsibilities.

Approx. 14 hours | **CO3** | **Understand · Apply**

Learning goals

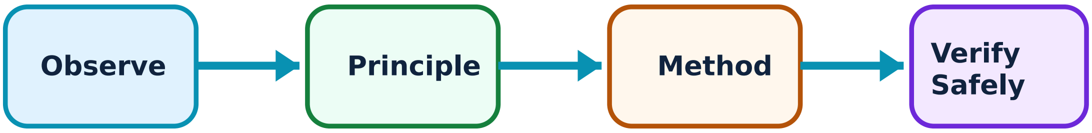
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Refrigerants, Controls and Applications: identify the system, state its principle, follow the correct method and verify the result safely.

1. Expansion devices and flow control−

Capillary tubes, automatic valves and thermostatic expansion valves regulate refrigerant flow and create the pressure drop feeding the evaporator. Their operation is linked to system capacity, superheat/control requirement, charge condition and clean flow paths.

Study and application method

Compare a capillary tube and thermostatic expansion valve by control principle, typical application, advantage, limitation and the observations needed before diagnosing poor performance.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare a capillary tube and thermostatic expansion valve by control principle, typical application, advantage, limitation and the observations needed before diagnosing poor performance.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Flow-control adjustment and troubleshooting can expose personnel to pressure and refrigerant. Only competent persons should perform service actions with the prescribed recovery equipment.

2. Refrigerant selection and applications−

Refrigerants are considered for thermodynamic behaviour, compatibility, safety class, environmental impact, operating pressure, efficiency and regulations. Refrigeration applications include food preservation, cold storage, ice production, water cooling and comfort conditioning.

Study and application method

Use a transparent selection checklist: cooling temperature, capacity, energy performance, toxicity/flammability class, environmental compliance, equipment compatibility, leak-management and end-of-life recovery.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a transparent selection checklist: cooling temperature, capacity, energy performance, toxicity/flammability class, environmental compliance, equipment compatibility, leak-management and end-of-life recovery.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Refrigerant rules and approved fluids change. Do not rely on historic examples for present-day selection or handling; consult current legal, manufacturer and institutional guidance.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Psychrometrics and Comfort Air Conditioning

Moist-air properties, psychrometric chart, heating/cooling/humidification/dehumidification/mixing processes, thermal comfort, simple cooling load and air-distribution awareness.

Approx. 16 hours **CO4** **Understand · Apply**

Learning goals

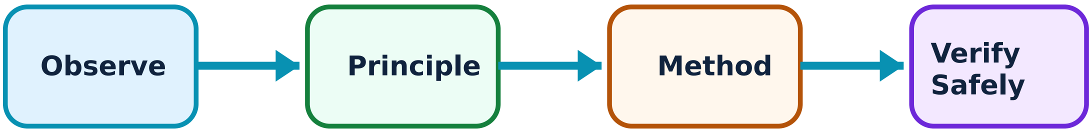
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Psychrometrics and Comfort Air Conditioning: identify the system, state its principle, follow the correct method and verify the result safely.

1. Moist-air properties and psychrometric processes−

Dry-bulb temperature, humidity ratio, relative humidity, wet-bulb temperature, dew point, enthalpy and specific volume describe moist air. Psychrometric charts represent these relationships and support study of sensible heating/cooling, humidification, dehumidification and mixing.

Study and application method

Locate supplied air states on a labelled chart or data table, state the process direction, identify which properties change and explain the physical conditioning action.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Locate supplied air states on a labelled chart or data table, state the process direction, identify which properties change and explain the physical conditioning action.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□□.

Common mistake / precaution: Psychrometric-chart readings are approximate and depend on pressure/altitude and data accuracy; state assumptions and do not treat a chart estimate as final design certification.

2. Comfort, cooling load and system view−

Comfort depends on temperature, humidity, air movement, activity, clothing, radiant conditions and indoor-air quality. Cooling-load studies account for envelope, occupants, lighting/equipment, ventilation/infiltration and process loads before selecting equipment and air-distribution arrangements.

Study and application method

Prepare a simple load inventory that separates sensible and latent contributors, records data source/assumption and identifies information still needed for a professional design.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a simple load inventory that separates sensible and latent contributors, records data source/assumption and identifies information still needed for a professional design.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Building HVAC design must follow applicable codes, energy requirements, fire/smoke provisions and qualified design review; avoid sizing equipment from a single simplified calculation.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Refrigeration Fundamentals and Cycles

Use the concept flow to revise observation, principle, method and verification.

Module 2: Absorption Systems and Refrigeration Equipment

Use the concept flow to revise observation, principle, method and verification.

Module 3: Refrigerants, Controls and Applications

Use the concept flow to revise observation, principle, method and verification.

Module 4: Psychrometrics and Comfort Air Conditioning

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Draw and annotate air-refrigeration and vapour-compression cycles, including component functions and energy transfers.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Complete a supplied P–h or state-point exercise with units, assumptions and a COP interpretation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a component inspection checklist for a simulated refrigeration plant that avoids unsafe service actions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Use a psychrometric chart or supplied property data to analyse a cooling-and-dehumidification process.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a preliminary cooling-load inventory for one room, separating known data from justified assumptions and information gaps.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Refrigeration Fundamentals and Cycles

Explain Refrigeration effect and performance and state one correct method, application or precaution.−

Refrigeration removes heat from a controlled low-temperature space and rejects it elsewhere. Refrigerating effect expresses useful cooling; coefficient of performance compares useful cooling to required work. A valid calculation needs a clear system boundary and consistent units.

Answer framework: For a supplied cycle sketch or state data, identify the refrigerated space, heat absorbed, work input, heat rejected and the appropriate COP expression before substituting values.

Important point: Do not use classroom COP calculations to predict an installed system without actual load, losses, controls, ambient conditions and component data.

Explain Air and vapour-compression cycles and state one correct method, application or precaution.−

Air refrigeration commonly uses reverse-Brayton principles. A simple vapour-compression system circulates refrigerant through compressor, condenser, expansion device and evaporator; phase change enables effective heat transfer. T–s and P–h diagrams help visualise state changes.

Answer framework: Trace each component in the cycle, label pressure/temperature trends and identify where work, heat rejection, throttling and refrigerating effect occur.

Important point: Pressure and temperature conditions can be hazardous. Never open, charge or modify a circuit as a learning exercise without authorisation and certified equipment.

Module 2 — Absorption Systems and Refrigeration Equipment

Explain Vapour-absorption refrigeration and state one correct method, application or precaution.−

Absorption systems use a heat-driven generator/absorber arrangement in place of a mechanical compressor for part of the compression function. Their application depends on available heat, working-pair choice, cooling-water needs, scale and efficiency considerations.

Answer framework: Draw the conceptual absorber–pump–generator–condenser–expansion–evaporator path and distinguish heat input, heat rejection, solution circulation and refrigerant circulation.

Important point: Working fluids, heat sources and pressurised equipment require system-specific knowledge; do not assume different absorption pairs have interchangeable hazards or maintenance needs.

Explain Major components and heat rejection and state one correct method, application or precaution.−

Compressors raise vapour pressure; condensers reject heat; expansion devices meter/pressure-reduce refrigerant; evaporators absorb heat. Air- and water-cooled condensers, cooling towers and evaporator types are selected according to duty, environment, water availability and maintenance capability.

Answer framework: For each component, state inlet/outlet condition trends, primary function, likely fault symptom and a non-invasive inspection item.

Important point: Electrical drives, moving fans, hot surfaces, water-treatment chemicals and high-pressure parts require lockout, guarding and approved maintenance practice.

Module 3 — Refrigerants, Controls and Applications

Explain Expansion devices and flow control and state one correct method, application or precaution.−

Capillary tubes, automatic valves and thermostatic expansion valves regulate refrigerant flow and create the pressure drop feeding the evaporator. Their operation is linked to system capacity, superheat/control requirement, charge condition and clean flow paths.

Answer framework: Compare a capillary tube and thermostatic expansion valve by control principle, typical application, advantage, limitation and the observations needed before diagnosing poor performance.

Important point: Flow-control adjustment and troubleshooting can expose personnel to pressure and refrigerant. Only competent persons should perform service actions with the prescribed recovery equipment.

Explain Refrigerant selection and applications and state one correct method, application or precaution.−

Refrigerants are considered for thermodynamic behaviour, compatibility, safety class, environmental impact, operating pressure, efficiency and regulations. Refrigeration applications include food preservation, cold storage, ice production, water cooling and comfort conditioning.

Answer framework: Use a transparent selection checklist: cooling temperature, capacity, energy performance, toxicity/flammability class, environmental compliance, equipment compatibility, leak-management and end-of-life recovery.

Important point: Refrigerant rules and approved fluids change. Do not rely on historic examples for present-day selection or handling; consult current legal, manufacturer and institutional guidance.

Module 4 — Psychrometrics and Comfort Air Conditioning

Explain Moist-air properties and psychrometric processes and state one correct method, application or precaution.−

Dry-bulb temperature, humidity ratio, relative humidity, wet-bulb temperature, dew point, enthalpy and specific volume describe moist air. Psychrometric charts represent these relationships and support study of sensible heating/cooling, humidification, dehumidification and mixing.

Answer framework: Locate supplied air states on a labelled chart or data table, state the process direction, identify which properties change and explain the physical conditioning action.

Important point: Psychrometric-chart readings are approximate and depend on pressure/altitude and data accuracy; state assumptions and do not treat a chart estimate as final design certification.

Explain Comfort, cooling load and system view and state one correct method, application or precaution.−

Comfort depends on temperature, humidity, air movement, activity, clothing, radiant conditions and indoor-air quality. Cooling-load studies account for envelope, occupants, lighting/equipment, ventilation/infiltration and process loads before selecting equipment and air-distribution arrangements.

Answer framework: Prepare a simple load inventory that separates sensible and latent contributors, records data source/assumption and identifies information still needed for a professional design.

Important point: Building HVAC design must follow applicable codes, energy requirements, fire/smoke provisions and qualified design review; avoid sizing equipment from a single simplified calculation.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Refrigeration Fundamentals and Cycles.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Absorption Systems and Refrigeration Equipment.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Refrigerants, Controls and Applications.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Psychrometrics and Comfort Air Conditioning.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Refrigeration effect, COP, units, air-refrigeration concept, reverse-Brayton/Bell–Coleman cycle and basic vapour-compression refrigeration..
2. Develop a complete answer or practical plan for: Vapour-absorption principle; compressors, condensers, evaporators, cooling towers and auxiliaries; construction, function and basic performance links..
3. Develop a complete answer or practical plan for: Expansion devices, refrigerant properties and designation, selection criteria, applications such as cold storage and domestic systems, environmental and safety responsibilities..
4. Develop a complete answer or practical plan for: Moist-air properties, psychrometric chart, heating/cooling/humidification/dehumidification/mixing processes, thermal comfort, simple cooling load and air-distribution awareness..

LAST-MILE REVISION

Quick Revision

Module 1: Refrigeration Fundamentals and Cycles

Refrigeration effect, COP, units, air-refrigeration concept, reverse-Brayton/Bell–Coleman cycle and basic vapour-compression refrigeration.

- Match each topic to CO1.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 2: Absorption Systems and Refrigeration Equipment

Vapour-absorption principle; compressors, condensers, evaporators, cooling towers and auxiliaries; construction, function and basic performance links.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Refrigerants, Controls and Applications

Expansion devices, refrigerant properties and designation, selection criteria, applications such as cold storage and domestic systems, environmental and safety responsibilities.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Psychrometrics and Comfort Air Conditioning

Moist-air properties, psychrometric chart, heating/cooling/humidification/dehumidification/mixing processes, thermal comfort, simple cooling load and air-distribution awareness.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Refrigeration effect and performance	Refrigeration removes heat from a controlled low-temperature space and rejects it elsewhere.
Air and vapour-compression cycles	Air refrigeration commonly uses reverse-Brayton principles.
Vapour-absorption refrigeration	Absorption systems use a heat-driven generator/absorber arrangement in place of a mechanical compressor for part of the compression function.
Major components and heat rejection	Compressors raise vapour pressure; condensers reject heat; expansion devices meter/pressure-reduce refrigerant; evaporators absorb heat.
Expansion devices and flow control	Capillary tubes, automatic valves and thermostatic expansion valves regulate refrigerant flow and create the pressure drop feeding the evaporator.
Refrigerant selection and applications	Refrigerants are considered for thermodynamic behaviour, compatibility, safety class, environmental impact, operating pressure, efficiency and regulations.
Moist-air properties and psychrometric processes	Dry-bulb temperature, humidity ratio, relative humidity, wet-bulb temperature, dew point, enthalpy and specific volume describe moist air.
Comfort, cooling load and system view	Comfort depends on temperature, humidity, air movement, activity, clothing, radiant conditions and indoor-air quality.

LEARNING RESOURCES

References

Recommended refrigeration and air-conditioning references

1. R. S. Khurmi and J. K. Gupta, A Textbook of Refrigeration and Air Conditioning.
2. C. P. Arora, Refrigeration and Air Conditioning.
3. W. F. Stoecker and J. W. Jones, Refrigeration and Air Conditioning.
4. ASHRAE Handbook—Fundamentals (consult current edition for professional practice).

Approved public refrigeration and air-conditioning sources

- https://www.ganeshpolytechnic.edu.in/images/upload/documents/395809245Refrigeration%20an conditioning%20%20By%20ER.%20Abhijit%20Dash.pdf
- https://nptel.ac.in/courses/112107208

These sources provide the verified study basis while official Course 6021T detail is unavailable; they do not replace applicable codes, manufacturer instructions, safe-work procedures or refrigerant regulations.